

Optimization Modeling Project

“Allocation of EV Charging Stations with DC Fast Chargers in Colorado Along Major Highways”

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Introduction:

Colorado is among the nation's leading states in electric vehicle (EV) adoption, with battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs) collectively accounting for over 25% of the vehicle market share (Resource 1). This rapid growth highlights the urgent need for a strategically planned and scalable public charging network capable of supporting both current EV owners and anticipated future growth.

The State of Colorado has established a comprehensive public charging strategy that addresses both highway corridor needs and urban charging demand (Resource 2). While this plan provides a strategic framework, there is an opportunity to further optimize the deployment of direct current fast charging (DCFC) stations at the county level. Refinement is especially important when considering gaps in the existing infrastructure, the distribution of electric vehicles, and projected growth in EV adoption over the coming years. With this information at hand our project focuses on optimizing the location per county of the new charging stations with the current 2025 budget for electrification of the highway corridor.

The primary objective of this project is to analyze county-level EV demand and charging infrastructure in Colorado and develop an optimization model to guide the deployment of new fast-charging stations along highways. By incorporating DC fast chargers into the model, we aim to maximize coverage, alleviate congestion at existing stations, and efficiently allocate financial resources efficiently within the available budget.

Our analysis combines multiple datasets, including existing public charging stations, county-level EV ownership, and corridor distances from state highway maps. Using these inputs, we estimate the demand as the difference between the current number of fast stations and the units required per county. Based on this demand, we formulated and solved an optimization model that identifies the optimal number of chargers to install in each county given the current \$57 million budget (Resource 7).

Statement of the problem and results:

The problem we address in our project is: how many new DC fast chargers should be installed in each Colorado county to maximize highway corridor coverage and meet projected EV demand, subject to the \$57.1 million statewide budget?

To answer this question, we developed a county-level optimization model that integrates spatial, demographic, and infrastructure data. The model uses several key elements derived from our datasets:

- Existing public charging infrastructure (from AFDC), cleaned and aggregated to the county level (Table 1 in Appendices);
- EV ownership per county, representing current demand and providing a basis for estimating future needs (Table 2 in Appendices);
- Highway corridor mileage per county, obtained from state mapping resources and used to define corridor-based charging requirements (Table 3 in Appendices);
- CDOT regional classification, allowing regional interpretation and consistency with state planning zones (Table 3 in Appendices);
- NEVI budget and cost parameters, including the total \$57.1 million allocation and an estimated cost of \$900,000 per new DC fast-charging station.

Based on these inputs, we estimated the number of chargers each county should have by applying spacing standards that reflect corridor traffic levels. Following state planning logic, Front Range counties (Adams, Arapahoe, Denver, Douglas, and Boulder) were assigned a higher-density requirement of one fast charger per 3 highway miles, while all other counties were assigned one charger per 10 highway miles. We then calculated demand as the difference between this required number of chargers and the number of existing DC fast ports in each county (Tables 4, 5 in Appendices). Counties with large gaps between required and existing infrastructure were assigned higher demand scores.

We formulated an optimization model that selects the number of new DC fast chargers to install in each county. The objective is to maximize total statewide demand satisfied, ensuring the new chargers reduce county-level shortages as efficiently as possible. The model is subject to the overall NEVI budget constraint, the unit cost per station, the region fairness (each region must get at least 10% of spend) and non-negativity/integer constraints for charger allocations.

The optimization results show that the highest-priority counties for new DC fast chargers are those with long stretches of highway, modest existing infrastructure, and rapidly growing EV ownership. Many rural corridor counties: including those in the Western Slope, Southern Colorado, and Eastern Plains, show substantial gaps between required and existing charger counts. These counties receive the largest recommended allocations because they are essential for long-distance EV travel but currently underserved (Tables 6 in Appendices).

Front Range urban counties already possess relatively dense networks of DC fast chargers, so the model assigns them fewer new stations unless their highway coverage requires additional support. Overall, the results confirm that Colorado can achieve its target deployment of approximately 63 new EV charging stations with DC fast charger while remaining within the \$57.1 million budget (total budget spent is \$56.7 million). The optimized allocation also aligns closely with NEVI corridor goals, improving statewide connectivity along I-25, I-70, US-50, and other major corridors.

This model provides a county-level, data-driven foundation for planning DC fast-charging infrastructure that complements the state's broader EV strategy. It highlights where investments will have the highest impact and offers a scalable framework for future expansion as EV adoption continues to increase.

Data Collection and Description:

For this project, we collected and processed multiple datasets to support the analysis and optimization of EV charging infrastructure in Colorado. The primary datasets include

information on existing EV charging stations, the number of electric vehicles (EVs) per county, the mapping of zip codes to counties and the corridor distance per county on minor highways.

The first dataset, containing existing EV charging stations for 2025 in Colorado, was downloaded from the Alternative Fuels Data Center (AFDC) (Resource 3). This dataset includes information about 2700 ev charging stations (addresses, zip codes, types of chargers, time and so on). However, it does not provide county-level information, which is necessary for our county-based analysis.

To address this, we collected a second dataset that maps all zip codes in Colorado to their corresponding counties. Using this mapping, we merged the charging stations dataset with the zip code dataset to assign each station to a county. During this process, some zip codes in the original datasets were incorrect or missing, so we performed data cleaning to correct these inconsistencies. Additionally, we removed unnecessary columns from the charging stations dataset to retain only the relevant information for our analysis. Then we aggregated the data to obtain the total number of Level 2 and DC fast chargers, number of charging stations, and total chargers for each county.

We also collected county-level data on the number of electric vehicles in Colorado. This dataset provides the total number of EVs for each county (210,152 registered EVs in Colorado) and allows us to estimate demand for charging infrastructure (Resource 4).

To account for corridor-based charging needs, we estimated corridor distances per county on minor highways using maps provided by the Colorado government (Resource 5). This information allows us to prioritize counties with highway corridors for new charging stations. Finally, to complete the county level information we used the CDOT region map to assign a region to each county (Resource 6).

After analyzing these datasets, we made an important modeling decision: we chose to focus exclusively on DC fast chargers. This decision aligns with the State of Colorado's strategy, which emphasizes the construction of DC fast-charging stations along major highways and in

key communities to support the development of statewide EV fast-charging corridors. Level 2 chargers, while useful for local charging, do not support long-distance mobility in the same way and are not the priority of the state's corridor-based plan. Level 1 Charging stations are also eliminated from the data because they are mostly home base due to the long time for charging.

We also incorporated cost and funding data from Colorado's National Electric Vehicle Infrastructure (NEVI) Program. The state has been allocated \$57.1 million new DC fast chargers, implying an average cost of about \$900,000 per new charging station with fast charger (resource 7, 8). This cost and budget information is used directly as constraints in our optimization model.

To represent corridor needs, we differentiated charger spacing by county. Front Range counties with higher traffic: Adams, Arapahoe, Denver, Douglas, and Boulder, were assigned one fast charger per 3 miles of highway. All other counties were assigned one charger per 10 miles, reflecting lower traffic and longer rural road segments. Integrating these NEVI program parameters ensures that our optimization results are realistic, policy-aligned, and our model focuses solely on allocating DC fast chargers along major highways.

Methodology and Model:

Data Selection:

The initial dataset contained all Colorado counties, but not all were appropriate candidates for charger deployment. After estimating corridor based charging demand and comparing it with the existing supply of DC fast chargers, we identified the subset of counties with unmet demand. Only these corridor counties were included in the optimization model. This prioritization ensures that investment is directed toward areas where new infrastructure would provide meaningful benefit and avoids allocating chargers to counties whose existing supply already meets estimated needs. Restricting the model to counties with unmet demand also improves computational efficiency by removing unnecessary decision variables.

Model:

The charger allocation problem is formulated as a mixed integer linear programming model. This is due to the fact that both the objective function and all constraints are linear expressions, while the decision variables must be whole values.

Decision Variables:

For each county (c), with unmet corridor based demand, the model defines a decision variable:

x_c = *how many DC fast chargers should be built in county c .*

The decision variables are restricted to being nonnegative integers in order to account for the fact that a fraction of a charger cannot be built. Thus, allocation of chargers must also be limited to a whole number.

Objective Function:

The objective of the model is to maximize satisfied demand per county. Only counties whose estimated demand exceeds their current supply are eligible to be allocated DC fast chargers under this model. A constant per-unit installation cost is a core assumption of this model which allows the objective to be represented as a linear expression.

Constraints:**Constraint 1: *Budget***

A budget constraint of \$57.1 million ensures that the total spending on charger installations does not exceed available resources. This enforces financial feasibility within the model.

Constraint 2: *Minimum and Maximum Allocation*

For each county, the number of chargers allocated is constrained to be at least zero and at most equal to the county's unmet demand. These bounds prevent over-allocation in areas where additional chargers are not needed and ensure decision variables remain within realistic limits.

Constraint 3: *Regional Fairness*

Each county with unmet demand is assigned to one of four broader regions. To avoid disproportionate investment in any single region, the model requires that each region receive at least 10% of total spending. This constraint promotes geographic equity and supports a more balanced statewide charging network.

Constraint 4: *County Fairness/ Minimum Coverage*

To ensure a basic level of service across the corridor network, the model mandates that each eligible corridor county receive at least one new charger. This prevents the solution from overlooking low demand areas and supports continuous EV mobility across the state's major travel routes.

Model Limitations:

Uniform Installation Cost:

The model assumes a constant installation cost for all chargers, regardless of county. In reality, installation costs vary considerably due to factors such as land and labor costs, permitting requirements, and the need for infrastructure upgrades. In addition, the model accounts for only the upfront installation cost of a DC fast charger. Real world deployments also incur ongoing maintenance and operational costs. By excluding these recurring and location specific costs, the model may underestimate the true long term financial requirements of charger deployment.

Generalizability:

The model relies on current estimates of demand, costs, and policy conditions. These parameters are likely to change over time as technology evolves, EV adoption increases, installation practices improve, or regulatory incentives shift. Because the model is calibrated to a specific set of present-day assumptions, its results may not generalize without adjustments to the parameters.

Over-Simplification:

The model treats each county as an independent unit and does not consider that chargers installed in one county may also serve drivers from neighboring counties. Moreover, the model specifies only how many chargers should be built in each county, not where within the country they should be located. As a result, the model may overlook opportunities to position chargers in locations that could efficiently serve multiple counties or enhance corridor connectivity.

Numerical Experiment and Results:

As a result of our numerical experimentation, the final model provides the most equitable and operationally efficient distribution of charging stations across Colorado counties. The results indicate that no county is substantially underserved, with the largest difference between demand and allocation being four stations in Washington County (Table 6 in Appendices). The model also responds appropriately to counties with unusually high demand, such as Yuma and Park, by allocating proportionally larger shares of the budget to those areas (Table 6 in Appendices). The cost parameters and county-level demand estimates used in the model were derived from ranges identified in preliminary research and therefore should be viewed as informed approximations rather than precise empirical values. Several constraints were designed in a similar way to reflect plausible state-level policy objectives, including the requirement that any county along a major highway be eligible for a minimum portion of the budget and receive at least one charging station when demand is present. Because these estimates and constraints are approximate and intentionally encourage broader coverage instead of concentration in high-demand areas, the model is well suited for numerical experimentation. Adjusting these assumptions allows us to evaluate the sensitivity of the allocation outcomes and to observe how different policy choices affect the optimal solution.

Higher-Than-Expected Demand Experiment:

In the first experiment, we investigated how the model responds when county-level demand is higher than initially estimated. To examine this, we increased demand estimates by 15 percent for every county. Somewhat unexpectedly, the resulting allocation became slightly less equitable than that of the base model. Several counties received only one charging station, while counties that initially exhibited higher demand experienced a sharper increase in allocated chargers that mirrors their elevated relative demand (Table 7 in Appendices). This behavior is likely driven by the multiplicative adjustment applied to all counties. Because each county's demand is scaled proportionally rather than increased by a constant increment, high-demand counties become disproportionately more influential in the optimization process, ultimately shifting the allocation toward those counties.

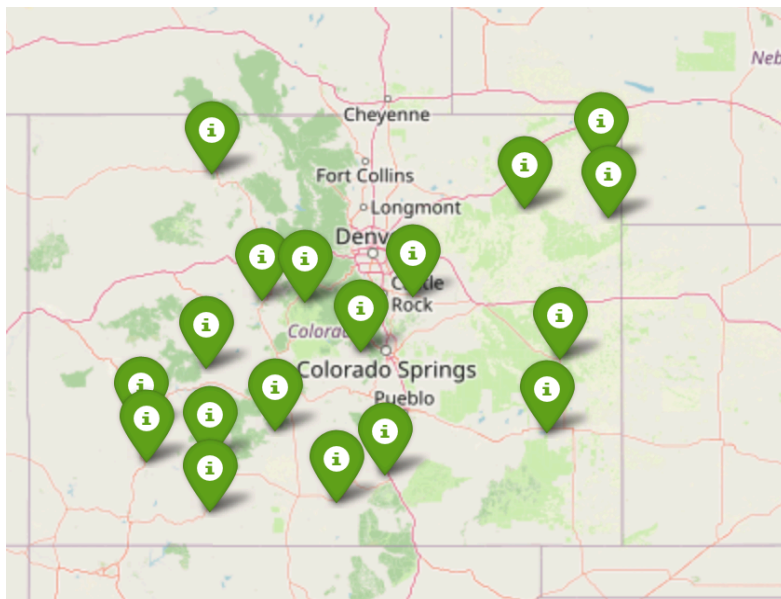
Higher Proportional Region Allocation Experiment:

A second experiment explored the effects of modifying the regional minimum-allocation constraint, which governs the proportion of the total budget assigned to each of the four regions. In the base model, this proportion was set to ten percent, a value that provided a reasonable balance between equitable geographic distribution and the need to serve areas with high demand. For the experimental model, we increased this proportion to twenty percent. Although the revised constraint improved geographic balance by distributing resources more evenly across counties, it simultaneously reduced the model's ability to meet demand in higher-demand areas (Table 8 in Appendices). For instance, Yuma County, which contains an important highway corridor and has a substantial demand of thirteen charging stations, was allocated only six stations under the more restrictive regional distribution requirement (Table 8 in Appendices). This illustrates the tradeoff inherent in imposing stronger equity-driven constraints, particularly when high-demand counties are concentrated in specific regions.

Changing the Minimum Allocation per County Experiment:

In a third experiment, we evaluated how the model responds to changes in the minimum number of charging stations allocated to each county. The final version of our model sets this minimum to one. This requirement ensures that every county, including those with demand as low as one station, receives at least some investment. To test the sensitivity of this constraint, we reduced the minimum allocation to zero, thereby removing the guarantee that all counties must receive at least one station. Under this modified specification, the model assigned no charging stations to Washington County and Bent County, despite Washington having a demand of five and Bent having a demand of four (Table 9 in Appendices). This observation highlights the importance of the minimum-allocation constraint in maintaining coverage across all counties. Without such a requirement, the optimization favors higher-demand areas to such an extent that meaningful gaps emerge in the charging network. From a policy perspective, ensuring that each county receives at least one station appears essential for achieving a coherent statewide charging network and for maintaining sufficient connectivity along major corridor routes.

Map of the Optimization Result:



Conclusions:

This optimization project has demonstrated to provide a balanced approach in allocating DC fast charger stations across Colorado counties, achieving both equity and operational efficiency in accordance with 2025 National Electric Vehicle Infrastructure Program Deployment Plan for Colorado (NEVI). The numerical experiments reveal several key insights about the model's behavior and tradeoffs inherent in public infrastructure planning.

The base model performs well in meeting its objective with no county substantially underserved. The model optimization responds to the high-demand counties as Yuma and Park when allocating larger shares of the budget, while still maintaining coverage across all counties thanks to the minimum allocation constraint.

The sensitivity analyses conducted through three distinct experiments illuminate critical policy considerations. First, when demand increased uniformly by 15%, the model shifted allocations to already high-demand counties decreasing equity to the other counties. This output shows the multiplicative effect when existing disparities are present and suggest that a demand base model can create several issues when allocating budget. The Second experiment, when we increased the equity constraint from 10% to 20%, improved the distribution balance, while compromising the needs of high-demand counties. Finally the elimination of the minimum one-station-per-county requirement created a gap in the allocation of budget to Washington and Bent counties decreasing the equity of distribution to all counties and in doing so reducing the overall network connectivity of the major highways that cross the state.

Recommendations:

1. **Maintaining the Minimum Allocation Constraint:** As we completed the sensitivity analysis the minimum requirement of one station per county constraint is essential for ensuring statewide network connectivity and preventing significant coverage gaps, as it is critical for long distance travel.

2. **Improvement in Demand Estimation Methodology:** a uniform demand per highway length can be improved if added more nuance to the demand as it could be determined based on its characteristics; urban or rural highways and the estimated current and future increase in demand of usage of each highway.
3. **Flexibility in implementation:** as we have allocated the budget it could be beneficial to have a phased approach when building the stations; it could be to start with larger distances between stations in highways with low demand in order to monitor the actual demand to those stretches of highway that are less used and adjust if necessary to other locations that in-real world performance are showing to have a higher demand.

Resources:

1. EV Market in Q3 2025 year
<https://colorado.auto/wp-content/uploads/2025/10/2025-Colorado-Auto-Outlook-Q3.pdf>,
2. Colorado charging infrastructure needs to reach electric vehicle goals
https://drive.google.com/file/d/1-Z_tqbx5nM5EF9TbIKTd8x-MWscFhlbw/view?usp=sharing
3. Existing EV charging stations in Colorado (resource of the dataset)
https://afdc.energy.gov/data_download
4. Colorado EV Registration Data by County
<https://www.atlasevhub.com/market-data/state-ev-registration-data/#data-format>
5. Corridor distance per county, on minor highways the data was estimated from the Colorado government map
https://gis.colorado.gov/viewer/?page=Map-View#widget_1068=layer_visibility:%7B%22widget_1068-dataSource_1%22%3A%7B%22widget_1068-dataSource_1-1860dc19607-layer-20%22%3Atrue%7D%7D.active_datasource_id:dataSource_1
6. Colorado Department of Transportation Region Map
<https://www.codot.gov/business/civilrights/assets/cdotregionmap-1.pdf>
7. The National Electric Vehicle Infrastructure (NEVI) Program Deployment Plan for Colorado
https://www.codot.gov/programs/innovativemobility/assets/colorado-2025-nevi-plan_final.pdf,
8. NEVI DC Fast Charging Station Total Project Cost
<https://www.paren.app/blog/nevi-dc-fast-charging-station-total-project-cost-averages-915-000>
9. National Electric Vehicle Infrastructure (NEVI) Project Planning Resource Map
<https://cdot.maps.arcgis.com/apps/instant/sidebar/index.html?appid=c0fc45199c064912>

[8797fc196994f8b9¢er=-105.8494;38.6784&level=7&hiddenLayers=185176ff318-layer-5;18516e05536-layer-3;19a0cc99f34-layer-14;19272be199b-layer-12](#)

10. Current Opened DCFC in Highway corridors. Colorado Energy Office

<https://energyoffice.colorado.gov/transportation/grants-incentives/ev-fast-charging-corridors>

Appendices:

Table 1: Current EV Charging Stations in Colorado in 2025 year by County (part of the table)

	A	B	C	D	E	F	G	H	I	J	K	L
1	Station Name	Street Address	City	ZIP	Access Days Time	EV Level1 EVSE Num	EV Level2 EVSE Num	EV DC Fast Count	Latitude	Longitude	ID	county
2	Denver International	8100 Tower Rd	Commerce City	80022	24 hours daily; pay k	18	9	1	39.843886	-104.771138	38522	Adams
3	New Belgium Brewin	500 Linden St	Fort Collins	80524	11am-8pm daily; for customer use only		4		40.593417	-105.0684	42529	Larimer
4	City of Boulder - Mur	5050 Pearl St	Boulder	80301	24 hours daily; empl	1	1		40.024578	-105.232194	42542	Boulder
5	City of Boulder - Part	1739 Broadway	Boulder	80302	Drivers must bring th	2	1		40.0148	-105.2792	42543	Boulder
6	Tynan's Nissan	5811 S College Ave	Fort Collins	80525	Dealership business hours		1		40.505604	-105.078041	43021	Larimer
7	Tremont Street Gara	400 15th St	Denver	80202	Garage business hours; pay lot		2		39.742113	-104.991416	43833	Denver
8	Tynan's Nissan	780 S Havana St	Aurora	80012	Dealership business hours		2	1	39.70234	-104.86539	46520	Arapahoe
9	Tynan's Nissan	780 S Havana St	Aurora	80012			1		39.70234	-104.86539	46521	Arapahoe
10	Boulder Nissan	2285 28th St	Boulder	80301	24 hours daily		3	1	40.025019	-105.258915	46522	Boulder
11	Larry H Miller Nissan	10030 E Arapahoe F	Centennial	80112	Dealership business hours		2	1	39.59426	-104.87253	46524	Arapahoe
12	Larry H Miller Nissan	10030 E Arapahoe F	Centennial	80112			1		39.59426	-104.87253	46525	Arapahoe
13	South Colorado Sprin	1333 S Academy Bl	Colorado Springs	80916	Dealership business hours		1	1	38.81236	-104.75642	46526	El Paso
14	South Colorado Sprin	1333 S Academy Bl	Colorado Springs	80916			1		38.81236	-104.75642	46527	El Paso
15	Woodmen Nissan	6840 Vincent Dr	Colorado Springs	80918	Dealership business hours		1	1	38.93113	-104.81125	46528	El Paso
16	Woodmen Nissan	6840 Vincent Dr	Colorado Springs	80918			1		38.93113	-104.81125	46529	El Paso
17	Nissan of Durango	20704 Highway 160	Durango	81303	Dealership business hours		1		37.269708	-107.907728	46532	La Plata
18	Nissan of Durango	20704 Highway 160	Durango	81303			1		37.269708	-107.907728	46533	La Plata
19	Greeley Nissan	2625 35th Ave	Greeley	80634	Dealership business hours		1	1	40.39443	-104.736034	46537	Weld
20	Empire Lakewood Ni	14707 W Colfax Ave	Lakewood	80401	Dealership business hours		1		39.732606	-105.163711	46538	Jefferson
21	Empire Lakewood Ni	14707 W Colfax Ave	Lakewood	80401			1		39.732606	-105.163711	46539	Jefferson
22	Empire Littleton Nis	5067 S Wadsworth B	Littleton	80123	Dealership business hours		1		39.626641	-105.090506	46541	Jefferson
23	Empire Littleton Nis	5067 S Wadsworth B	Littleton	80123			1	2	39.626641	-105.090506	46542	Jefferson
24	Valley Nissan	1005 Ken Pratt Blvd	Longmont	80501	Dealership business hours		1	1	40.151989	-105.108644	46543	Boulder
25	Valley Nissan	1005 Ken Pratt Blvd	Longmont	80501			1		40.151989	-105.108644	46544	Boulder

Resource: This table was created be us (final_ev_stations.csv)

Table 2: The Number of Electric Vehicles in Colorado by County in 2025 year (part of the table)

	A	B
1	county	ev cars
2	Adams	12251
3	Alamosa	94
4	Arapahoe	24377
5	Archuleta	164
6	Baca	10
7	Bent	12
8	Boulder	23092
9	Broomfield	4894
10	Chaffee	468
11	Cheyenne	16
12	Clear Creek	138
13	Conejos	30
14	Costilla	38
15	Crowley	18

Resource: This table was created by us (num_ev_incounties.csv)

Table 3: Highway Corridor Mileage per County in Colorado with Regions (part of the table)

	A	B	C	D	E
1	Highway	County	aprox mileage	CDOT Region	Region
2	I-25	Las Animas	39	Southeast	2
3	I-25	Huerfano	30	Southeast	2
4	I-25	Pueblo	47	Southeast	2
5	I-25	El Paso	47	Southeast	2
6	I-25	Douglas	32	Central	1
7	I-25	Arapahoe	40	Central	1
8	I-25	Denver	31	Central	1
9	I-25	Adams	65	Central	1
10	I-25	Broomfield	4	Central	1
11	I-70	Mesa	65	Northwest	3
12	I-70	Garfield	65	Northwest	3
13	I-70	Eagle	60	Northwest	3
14	I-70	Clear Creek	33	Central	1
15	I-70	Jefferson	24	Central	1

Resource: This table was created by us (highway_county.csv)

Table 4: The Number of Level 2 and DC Fast Chargers by County (part of the table)

	A	B	C	D	E
1	county	total_level2_charger	total_dc_fast_charger	num_charging_station	total_chargers
2	Adams	340	95	164	435
3	Alamosa	12	12	11	24
4	Arapahoe	381	113	168	494
5	Archuleta	10	3	8	13
6	Bent	1	0	1	1
7	Boulder	787	88	362	875
8	Broomfield	116	26	68	142
9	Chaffee	36	27	21	63
10	Cheyenne	0	8	1	8
11	Clear Creek	17	30	19	47
12	Costilla	4	0	2	4
13	Custer	7	1	5	8
14	Delta	14	6	12	20
15	Denver	1231	169	425	1400
16	Dolores	3	0	2	3
17	Douglas	175	68	80	243
18	Eagle	247	24	120	271
19	El Paso	196	90	113	286
20	Elbert	4	0	2	4
21	Fremont	19	10	12	29

Resource: This table was created by us (charging_stations_per_county.csv)

Table 5: The Number of Needed Fast Charging Stations by County

	A	B
1	County	Needed
2	Archuleta	2
3	Bent	4
4	Cheyenne	3
5	Costilla	4
6	Elbert	6
7	Gunnison	3
8	Huerfano	5
9	Lake	5
10	Mineral	2
11	Ouray	2
12	Park	6
13	Phillips	3
14	Routt	2
15	Saguache	4
16	San Juan	3
17	Teller	1
18	Washington	5
19	Yuma	13

Resource: This table was created by us (needed_fastchargers.csv)

Table 6: Result of Optimization Model

	A	B	C	D
1	County	Demand	Optimized Chargers	Total Cost
2	Archuleta	2	2	1800000
3	Bent	4	4	3600000
4	Cheyenne	3	1	900000
5	Costilla	4	4	3600000
6	Elbert	6	6	5400000
7	Gunnison	3	3	2700000
8	Huerfano	5	5	4500000
9	Lake	5	5	4500000
10	Mineral	2	2	1800000
11	Ouray	2	2	1800000
12	Park	6	6	5400000
13	Phillips	3	1	900000
14	Routt	2	2	1800000
15	Saguache	4	4	3600000
16	San Juan	3	3	2700000
17	Teller	1	1	900000
18	Washington	5	1	900000
19	Yuma	13	11	9900000

Resource: This table was created by us (results_opt.csv)

Table 7: Results of Numerical Experiment 1 “Higher than Expected Demand”

	A	B	C	D
1	County	Demand	Optimized Chargers	Total Cost
2	Archuleta	3	1	900000
3	Bent	5	1	900000
4	Cheyenne	4	1	900000
5	Costilla	5	1	900000
6	Elbert	7	7	6300000
7	Gunnison	4	4	3600000
8	Huerfano	6	1	900000
9	Lake	6	6	5400000
10	Mineral	3	1	900000
11	Ouray	3	3	2700000
12	Park	7	7	6300000
13	Phillips	4	1	900000
14	Routt	3	3	2700000
15	Saguache	5	5	4500000
16	San Juan	4	4	3600000
17	Teller	2	2	1800000
18	Washington	6	1	900000
19	Yuma	15	14	12600000

Resource: This table was created by us (experiment1.csv)

Table 8: Results of Numerical Experiment 2 “Higher Proportional Regional Allocation”

	A	B	C	D
1	County	Demand	Optimized Chargers	Total Cost
2	Archuleta	2	1	900000
3	Bent	4	1	900000
4	Cheyenne	3	1	900000
5	Costilla	4	1	900000
6	Elbert	6	6	5400000
7	Gunnison	3	3	2700000
8	Huerfano	5	5	4500000
9	Lake	5	5	4500000
10	Mineral	2	1	900000
11	Ouray	2	2	1800000
12	Park	6	6	5400000
13	Phillips	3	1	900000
14	Routt	2	2	1800000
15	Saguache	4	4	3600000
16	San Juan	3	3	2700000
17	Teller	1	1	900000
18	Washington	5	1	900000
19	Yuma	13	6	5400000

Resource: This table was created by us (experiment2.csv)

Table 9: Results of Numerical Experiment 3 “Minimum Allocation per County reduced to Zero”

	A	B	C	D
1	County	Demand	Optimized Chargers	Total Cost
2	Archuleta	2	2	1800000
3	Bent	4	0	0
4	Cheyenne	3	3	2700000
5	Costilla	4	4	3600000
6	Elbert	6	6	5400000
7	Gunnison	3	3	2700000
8	Huerfano	5	5	4500000
9	Lake	5	5	4500000
10	Mineral	2	2	1800000
11	Ouray	2	2	1800000
12	Park	6	6	5400000
13	Phillips	3	3	2700000
14	Routt	2	2	1800000
15	Saguache	4	4	3600000
16	San Juan	3	3	2700000
17	Teller	1	1	900000
18	Washington	5	0	0
19	Yuma	13	12	10800000

Resource: This table was created by us (experiment3.csv)

Image 1: Current Opened DCFC in Highway corridors (Resource 10)

